

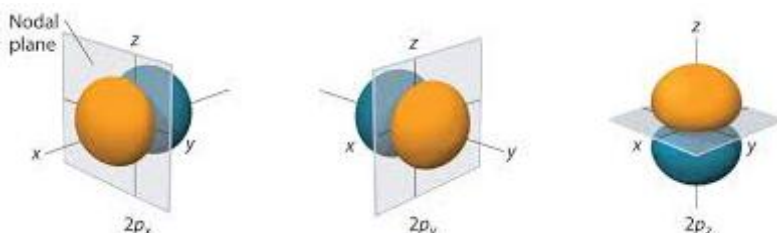
### Schrodinger Wave Mechanical model

- After the failure of Bohr atomic theory but its ability to explain the atomic spectra a need was felt for the new model that could incorporate, the concept of stationary orbit, de Broglie concept, Heisenberg uncertainty principle. The concept that incorporate above facts is called quantum mechanics of the atomic model or wave mechanical model.
- Ervin Schrodinger developed a model which is based on the particle and wave nature of the electron, known as Wave Mechanical Model of atom.
- Schrodinger applied the equation to the hydrogen atom and predicted many of its properties with remarkable accuracy. The differential wave equation is as follows:

$$\frac{\delta^2 \psi}{\delta x^2} + \frac{\delta^2 \psi}{\delta y^2} + \frac{\delta^2 \psi}{\delta z^2} + \frac{8\pi^2 m}{h^2} (E - V) \psi = 0$$

where m is mass of electron,  $\psi$  is wave function, E is total energy of electron, V is potential energy and h is Planck's constant.

- The equation determines the behavior of the wave function ( $\psi$ ) that describes the wave like properties of subatomic system. It is solved to find the different energy levels of the system.
- An atomic orbital is a one electron wave function ( $\psi$ ) obtained from the solution of the Schrodinger equation. The orbital wave function  $\psi$  has no physical significance but its square ( $\psi^2$ ) has a physical significance it measures the electron probability density at a point in an atom.
- When we solve the Schrodinger equation, it is observed that for some region of space, the value of  $\psi$  is positive and for other, it is negative. But the probability must be positive, so it is proper to use  $\psi^2$  in place of  $\psi$ .
- As a consequence of the Uncertainty Principle, the idea of an electron orbit is lost; it is replaced with a description of where the electron is most likely to be found. This total picture of the probability of finding an electron at various points in space is called an "orbital".
- The Schrodinger equation is said to have been solved for a particular atomic system. The details of, how this is done, are beyond the syllabus, but the consequences of its solution are extremely important to us.
- The important point of the solution of this equation is that it provides a set of numbers called quantum numbers. Quantum numbers are required to describe the distribution of electrons in atoms. Quantum numbers derived from the solution of Schrodinger equation are called principal quantum number (n), azimuthal quantum number (l) and magnetic quantum number (m). These quantum numbers are used to describe the atomic orbitals.
- Orbital** : The locations in space at which the probability of finding the electron is maximum.
- Node and Nodal Plane** : Node represents the region where probability of finding an electron is zero (i.e.,  $\psi$  and  $\psi^2 = 0$ ). Similarly nodal plane represents the plane having zero probability of finding electron.



- Nodes are of two types :**

- (a) Radial node : A radial node is the spherical region around nucleus having  $\psi$  and  $\psi^2$  equal to zero.
- (b) Angular node or Nodal plane : In this region (plane or surface),  $\psi$  and  $\psi^2$  equal to zero.

- An orbital having higher number of nodes has more energy.

- Calculation of number of nodes :** Radial nodes =  $n - \ell - 1$ ,  
Angular nodes =  $\ell$ ,  
Total nodes =  $n - 1$

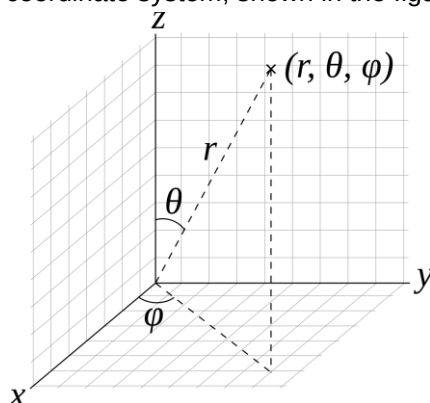
where  $n$  and  $\ell$  are principal and azimuthal quantum numbers.

E.g. In 3p-orbital, Radial nodes =  $3 - 1 - 1 = 1$  ( $= n - \ell - 1$ )  
Angular nodes =  $1$  ( $= \ell$ )  
Total nodes =  $2$  (one radial, one angular)

### Pictorial Representation of Atomic orbitals

To make the concept of an orbital more meaningful, it is helpful to examine the actual solution of the wave function for the one-electron atom.

- Because of the spherical symmetry of the atom, the wave functions are most simply expressed in terms of a spherical polar-coordinate system, shown in the figure, which has its orbit at the nucleus.



radial distance =  $r$ ,  
azimuthal angle =  $\theta$ ,  
and polar angle =  $\phi$ .  
 $x = r \sin \theta \cos \phi$   
 $y = r \sin \theta \sin \phi$   
 $z = r \cos \theta$

- It is found that the wave functions can be expressed as the product of two functions, one of which (the “angular part”  $X$ ) depends only the angle  $\theta$  and  $\phi$ , the other of which (the “radial part”  $R$ ) depends only on the distance from the nucleus.

- Orbital wave function  $\psi$  is the product of two functions, radial function  $\psi(r)$  and angular function  $\psi(\theta, \phi)$  i.e.,

$$\psi = \psi(r) \cdot \psi(\theta, \phi) \quad [ \psi(r) = R ; \quad \psi(\theta, \phi) = X ]$$

$$\text{Thus we have} \quad \psi(r, \theta, \phi) = R(r) \cdot X(\theta, \phi)$$

$\psi(r)$  is the function of distance ( $r$ ) from the nucleus while  $\psi(\theta, \phi)$  is the function of two spherical co-ordinates  $\theta$  and  $\phi$  as shown in above figure.

- $\psi(r)$  i.e. radial part of wave function depends upon quantum number ( $n$ ) and ( $\ell$ ) and decides the size of an orbital.
- Angular part of wave function  $\psi(\theta - \phi)$ , depends upon quantum numbers ( $\ell$ ) and ( $m$ ) and describes the shape of orbital
- Angular and radial parts of hydrogen atom wave functions

#### Angular part $X(\theta, \phi)$

$$X(s) = \left( \frac{1}{4\pi} \right)^{1/2}$$

$$X(p_x) = \left( \frac{3}{4\pi} \right)^{1/2} \sin \theta \cos \phi$$

#### Radial part $R_{n,\ell}(r)$

$$R(1s) = 2 \left( \frac{z}{a_0} \right)^{3/2} e^{-\sigma/2}$$

$$R(2s) = \frac{1}{2\sqrt{2}} \left( \frac{z}{a_0} \right)^{3/2} (2 - \sigma) e^{-\sigma/2}$$

$$R(2p) = \frac{1}{2\sqrt{6}} \left( \frac{z}{a_0} \right)^{3/2} \sigma e^{-\sigma/2}$$

$$R(3s) = \frac{1}{9\sqrt{3}} \left( \frac{z}{a_0} \right)^{3/2} (6 - 6\sigma + \sigma^2) e^{-\sigma/2}$$

$$R(3p) = \frac{1}{9\sqrt{6}} \left( \frac{z}{a_0} \right)^{3/2} (4 - \sigma) \sigma e^{-\sigma/2}$$

$$R(3d) = \frac{1}{9\sqrt{30}} \left( \frac{z}{a_0} \right)^{3/2} \sigma^2 e^{-\sigma/2}$$

$$\sigma = \frac{2Zr}{na_0} \quad ; \quad a_0 = \frac{h^2}{4\pi^2 m e^2}$$

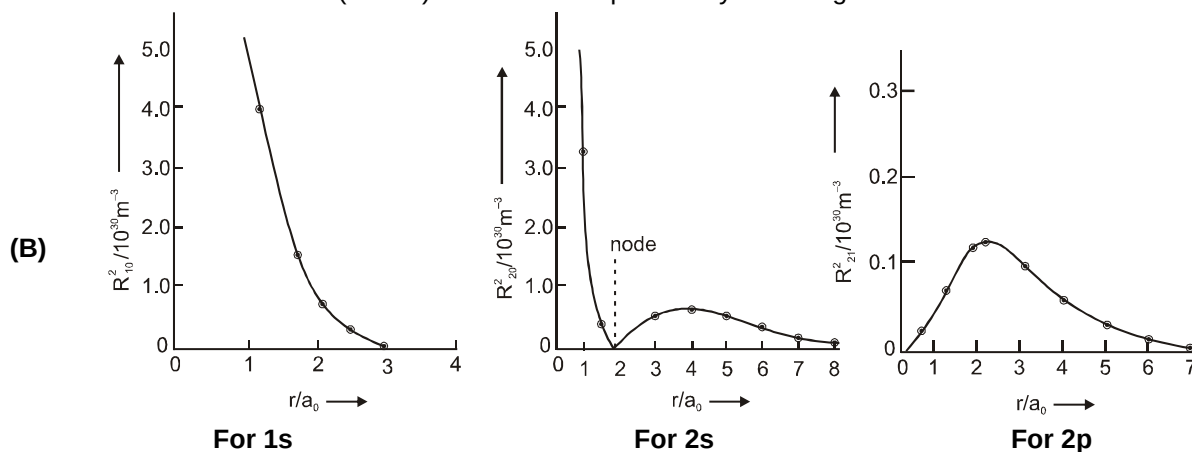
$$\sigma = \frac{2Zr}{na_0} \quad ; \quad a_0 = \frac{h^2}{4\pi^2 m e^2}$$

- For the sake of convenience the  $\psi(r)$  vs.  $r$  and  $\psi(\theta, \phi)$  vs. angle are plotted separately.

- 
- Figure 1 consists of three subplots labeled (A), (B), and (C), each showing the radial probability distribution  $R(r)$  for different atomic orbitals. The x-axis for all plots is  $r/a_0$ .
- (A) For 1s:** The y-axis is  $R^{1.0}/10^{15} \text{m}^{-1.5}$ . The curve starts at a maximum value of approximately 5.0 at  $r=0$  and decreases monotonically, approaching zero as  $r$  increases to 4.
  - (B) For 2s:** The y-axis is  $R^{2.0}/10^{15} \text{m}^{-1.5}$ . The curve starts at a maximum value of approximately 1.6 at  $r=0$ , decreases to a node (where  $R=0$ ) at  $r \approx 1.0$ , and then increases to a local maximum of approximately 0.1 at  $r \approx 4.0$  before decreasing again.
  - (C) For 2p:** The y-axis is  $R^{2.1}/10^{15} \text{m}^{-3.5}$ . The curve starts at zero at  $r=0$ , increases to a maximum value of approximately 0.4 at  $r \approx 2.1$ , and then decreases as  $r$  increases to 7.

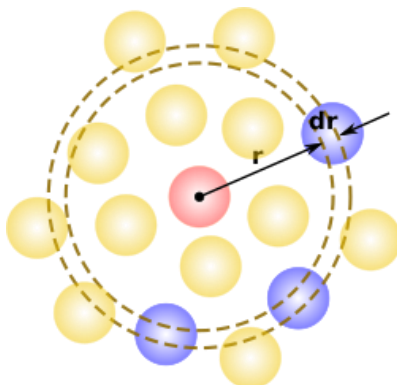
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- (ii) **Radial Probability density ( $R^2$ ) [Fig. (B)]** : The radial density  $R^2$  gives the probability density of finding the electron at a point along a particular radius line. Plots in fig. (B) give useful information about probability density or relative electron density at a point as a function of radius. It may be noted that for s-orbitals the maximum electron density is at the nucleus while all other orbitals have zero electron density at the nucleus. Its zero value ( $R^2 = 0$ ) indicates zero probability of finding an electron.

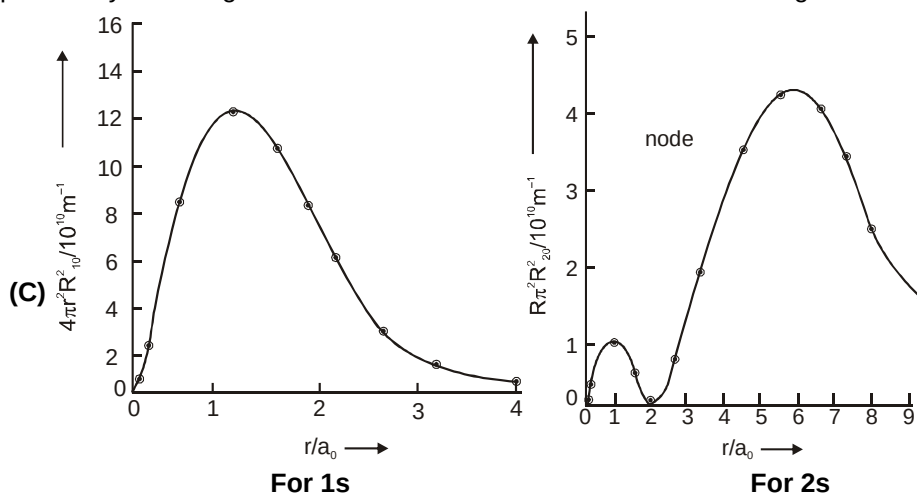


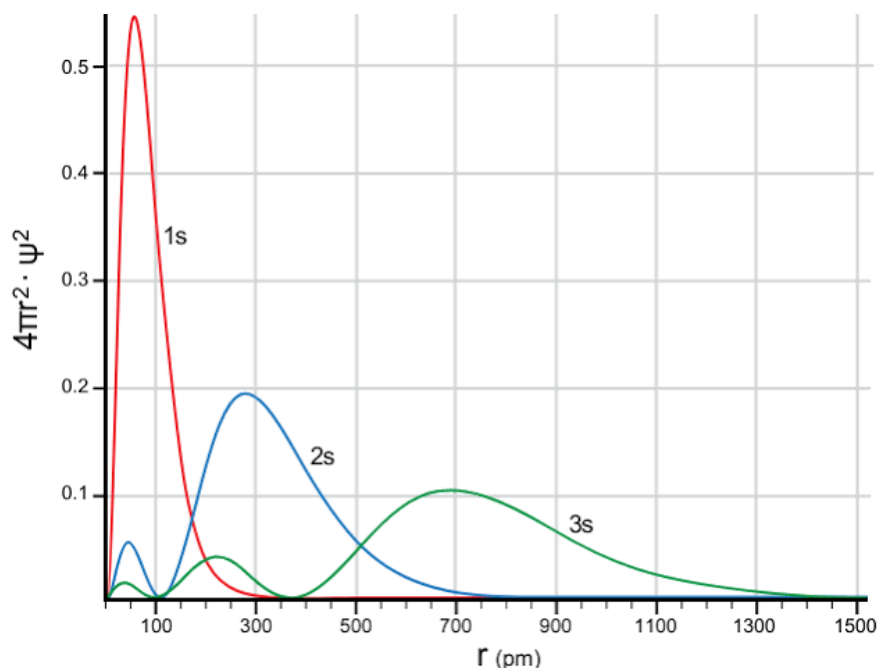
- (iii) **Radial probability function  $4\pi r^2 R^2$  [Fig. (C)]** :

- Radial Probability = Radial Probability Density  $\times$  Volume**  
Since the atoms have spherical symmetry, It is more useful to discuss the probability of finding the electron in a spherical shell between the spheres of radius  $(r + dr)$  and  $r$ . The volume of the shell is equal to  $\frac{4}{3}\pi(r + dr)^3 - \frac{4}{3}\pi r^3 = 4\pi r^2 dr$ .
- $\psi^2 dV$  (or  $\psi^2 \cdot 4\pi r^2 dr$ ) represents the probability for finding electron in a small volume  $dV$  surrounding the nucleus.



This probability which is independent of direction is called radial probability and is equal to  $[4\pi r^2 dr R^2]$ . It gives the probability of finding the electron at a distance  $r$  from the nucleus regardless of direction.





## PRACTICES PROBLEMS

- The quantum number not obtained from the Schrodinger's wave equation is :  
 (A)  $n$  (B)  $\ell$  (C)  $m$  (D)  $s$
- Solution of Schrodinger equation gives three quantum number ( $n$ ,  $\ell$ ,  $m$ ) as  $\psi_{n\ell m}$ . Which orbital is represented by the complete wave function  $\psi_{400}$  :  
 (A) 4d (B) 3d (C) 4p (D) 4s
- In  $\psi_{321}$ , the sum of orbital angular momentum, number of spherical nodes and number of angular nodes is :  
 (A)  $\frac{\sqrt{6}h + 4\pi}{2\pi}$  (B)  $\frac{\sqrt{6}h + 2\pi}{2\pi}$  (C)  $\frac{\sqrt{8}h + 4\pi}{2\pi}$  (D)  $\frac{\sqrt{8}h + 6\pi}{2\pi}$

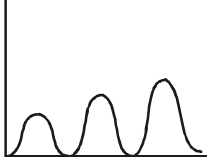
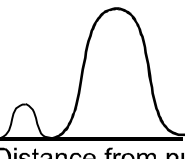
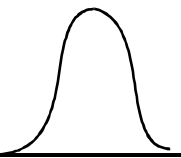
- For 3s-orbital of hydrogen atom, the normalised wave function is given by

$$\psi_{3s} = \frac{1}{18\sqrt{3\pi}} \left( \frac{1}{a_0} \right)^{3/2} \left[ 6 - \frac{4r}{a_0} + \frac{4r^2}{9a_0^2} \right] e^{-r/3a_0}$$

The above mentioned orbital (3s) has two nodes at  $1.9a_0$  and  $xa_0$ . Find the value of  $x$ .

- (A) 6.4 (B) 7.1 (C) 2.5 (D) 5.6
- \* Select the correct statement (s) :  
 (A) Radial function  $[R(r)]$  a part of wave function is dependent on quantum number  $n$  only  
 (B) Angular function depends only on the direction and is independent to the distance from the nucleus  
 (C)  $\psi^2(r, \theta, \phi)$  is the probability density of finding the electron at a particular point in space  
 (D) Radial distribution function ( $4\pi r^2 R^2$ ) gives the probability of the electron being present at a distance  $r$  from the nucleus

6. Find :
- (A) The number of radial nodes of 5s atomic orbital  
 (B) The number of angular nodes of  $3d_{yz}$  atomic orbital  
 (C) The sum of angular nodes and radial nodes of  $4d_{xy}$  atomic orbital  
 (D) The number of angular nodes of 3p atomic orbital
7. **Column-I** and **Column-II** contains four entries each. Entries of **Column-I** are to be matched with some entries of **Column-II**. One or more than one entries of **Column-I** may have the matching with the same entries of **Column-II**.
- |     | Column-I                                  |     | Column-II      |
|-----|-------------------------------------------|-----|----------------|
| (A) | The orbital which has two angular nodes   | (p) | $3d_{x^2-y^2}$ |
| (B) | The orbital which has one angular node    | (q) | $3p_x$         |
| (C) | The orbital without an angular node       | (r) | 4f             |
| (D) | The orbital which has three angular nodes | (s) | 3s             |
8. **Column-I** and **Column-II** contains four entries each. Entries of **Column-I** are to be matched with some entries of **Column-II**. One or more than one entries of **Column-I** may have the matching with the same entries of **Column-II**.

|     | Column-I |     | Column-II                                                                                                                    |
|-----|----------|-----|------------------------------------------------------------------------------------------------------------------------------|
| (A) | 3s       | (p) | $4\pi r^2 R^2$ <br>Distance from nucleus   |
| (B) | 3p       | (q) | $4\pi r^2 R^2$ <br>Distance from nucleus |
| (C) | 3d       | (r) | $4\pi r^2 R^2$ <br>Distance from nucleus |
| (D) | 2p       | (s) | $4\pi r^2 R^2$ <br>Distance from nucleus  |

## ANSWER KEY

- |                                                  |                                                  |        |        |
|--------------------------------------------------|--------------------------------------------------|--------|--------|
| 1. (D)                                           | 2. (D)                                           | 3. (A) | 4. (B) |
| 5. (B,C,D)                                       | 6. A - 4 ; B - 2 ; C - 3 ; D - 1                 |        |        |
| 7. (A) - (p) ; (B) - (q) ; (C) - (s) ; (D) - (r) | 8. (A) - (q) ; (B) - (r) ; (C) - (s) ; (D) - (p) |        |        |